Salient Is Not Enough: Keeping Every Time Anchor for Temporal Grounding under a Visual-Token Budget

Anonymous CVPR submission

Paper ID *****

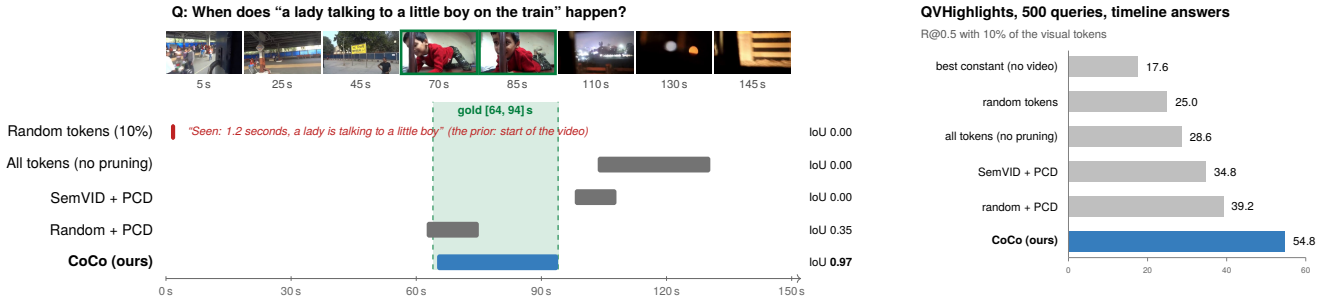


Figure 1. **Temporal grounding from a pruned video memory.** Left: a QVHighlights query, with the intervals predicted by each method at 10% of the visual tokens (real outputs). With random pruning the model falls back on a video-free prior and answers “1.2 seconds”; QA-oriented pruning and the uncompressed model pick a later, wrong scene. CoCo keeps every frame anchored in time and contrasts the answer against the model’s no-video prediction, and recovers the interval (IoU 0.97). Right: R@0.5 over 500 queries when every method answers with a timeline of observations, the format of grounded QA. In this format the prior writes many of the times and CoCo is far above the uncompressed model and the best video-free constant. With a one-line answer the uncompressed model is much stronger, and Coverage alone matches it with a quarter of the tokens (Sec. 6.2).

Abstract

Video LLMs are increasingly asked not only what *happened but* when*: a grounded question such as “how long did I use the paper cutter?” needs an answer and the time interval of its evidence. To keep long videos tractable, the visual memory of these models is pruned to a small fraction of its tokens, and published pruners lose little multiple-choice accuracy at 10% of the tokens. We show that temporal grounding needs something different:* pruning for QA keeps what is salient, but grounding needs what is everywhere. *A video LLM reads time from timestamps that survive only together with visual tokens of their frame, so pruners that select by saliency, uniqueness or text similarity empty frames, delete their timestamps, and fall* below random token selection *on grounding. When the answer lists observed moments before the interval, the gaps are filled by a video-free prior: asked with no video at all, the model still starts its intervals at the beginning of the window. We propose CoCo, a training-free method with two parts.* Coverage *gives every frame an equal token quota, chosen per layer by the actual question’s attention, so that every time anchor survives.* Prior-contrastive decoding *(PCD) writes each time value against the model’s own no-video prediction. The two parts behave differently across answer formats, which we study directly. Under the standard one-line grounding prompt, Coverage alone matches the uncompressed model with a quarter of the visual tokens on a general and a grounding-tuned model of the same family (QVHighlights, 61.7* vs. *60.5 and 67.3* vs. *67.1 mIoU) and beats random selection (+3.5 and +2.0), while PCD has no effect. In reasoning-style grounded QA (S-EMBER), PCD is what matters: with 10% of the tokens CoCo improves on the uncompressed model by +11.2 and +10.7 mIoU under the same prompt. Asking for a timeline of observations instead of a single interval costs single-moment grounding about 30 mIoU, so the answer format is a first-order variable in evaluating compressed memories. We evaluate every method against video-free floors and report where the method does not help: on short, densely sampled videos (Charades-STA, NExT-GQA), Coverage keeps the answers but not more of the interval.*

1. Introduction

Ask a video assistant "how long did I use the paper cutter?" and a useful answer has two parts: a duration and the interval it comes from. Such *grounded* questions are central to egocentric assistants, video search and step-by-step guidance. They are hard for the same reason they are useful: the model must know *when* things happened across the whole window, including the moments just before and after the event.

At the same time, video LLMs pay for every frame. Qwen3-VL [35] spends about 330 visual tokens per frame (660 per pair of frames), so 64 frames already cost about 21,000 tokens, and long or streaming video multiplies this. Pruning the visual tokens to a small budget is therefore the default, and a fast-growing literature reports that 10–25% of the tokens preserve multiple-choice accuracy [14, 17, 25, 33, 38, 41]. This paper asks what pruning does to *temporal grounding*. Our answer is a single principle with two consequences: **pruning for QA keeps what is salient, but grounding needs what is everywhere.** A QA answer can be read from a few informative frames; an interval can only be read from a timeline, and a timeline needs an anchor in every part of the window.

Observation 1: QA pruners lose the anchors. Video LLMs such as Qwen3-VL report time by reading timestamp tokens interleaved with the visual tokens. A frame whose visual tokens are all removed loses its timestamp, or keeps a timestamp with nothing to show. Pruners designed for QA remove frames that look redundant, unremarkable or unrelated to the words of the question, and those frames are exactly what makes a timeline usable. On S-EMBER grounded QA, all four recent pruners we evaluate fall below *random* token selection at the same budget, and the same holds on QVHighlights and Charades-STA (Sec. 6.3). Random selection is a strong baseline for grounding because it keeps a few tokens on every frame, and one of the published scores improves by 5.3^{*} mIoU simply by being ranked within each frame instead of globally (Sec. 3.2).

Observation 2: where the memory is silent, a prior answers. Grounded QA invites a reasoning-style answer: the model lists what it saw and when, then states the interval. Asked this way with no video at all, the model still produces confident intervals, and 90% of them start in the first 5% of the window. A pruned memory shows the same habit: 30–56% of predicted intervals start in the first 5% of the window, against 16% of the gold intervals, and removing the frames that contain the evidence raises this share from 37% to 51% (Sec. 3.3). The prior is therefore not a property of a particular pruner; it is what the decoder writes where the memory has nothing to say.

Observation 3: the answer format decides which failure dominates. The two observations call for two remedies, and they do not matter equally everywhere. On the same QVHighlights queries and the same memory, asking for a timeline of observations instead of a single interval costs about 30 mIoU (Sec. 6.2). Under the one-line format, keeping anchors is what helps and contrastive decoding does nothing; under the timeline format, the video-free prior writes many of the times, and contrasting it away is what helps. We therefore treat the answer format as an experimental variable, not a fixed detail of the protocol.

Observation 4: benchmark floors are high. Grounding benchmarks have strong positional regularities [34]. On S-EMBER, where questions are asked mid-stream and two thirds of the gold intervals end within the last fifth of the window, the best *constant* interval $[0.15q, q]$ reaches 36.9 mIoU, more than the uncompressed model answering in the timeline format (28.0). Following Otani et al. [34], we report video-free floors for every method (Sec. 3.4).

We propose CoCo (Coverage + PCD), a training-free method with one component for each failure (Fig. 3):

- **Coverage** (Sec. 4.1) keeps the anchors. After the frames are encoded, the KV cache is pruned layer by layer. Every frame keeps an equal share of the layer's budget, and within each frame the tokens that the *actual question* attends to most are kept. Because a timestamp survives exactly when one of its frame's visual tokens survives, the equal quota keeps every time anchor, and the question decides what each anchor shows.
- **Prior-contrastive decoding (PCD)** (Sec. 4.2) removes the prior where it writes the answer. Each time value is chosen by contrasting the model's prediction with its prediction from a video-free memory, the content-free contrast of contextual calibration and context-aware decoding [39, 49] applied only to time values, with a strength that fades where the model is already confident and an *answer reserve* (Sec. 4.2) that guarantees the interval is written.

Neither operator is new in isolation: per-frame budgets appear in VidCom^2, KTV and AdaTP [1, 12, 33], question-attention KV eviction in SnapKV and PruneVid [25, 30], and contrast against a content-free or degraded input in contrastive decoding, classifier-free guidance and VCD [18, 28, 29, 37, 39]. Our contributions are to show *why* temporal grounding needs each of them, *when* each one matters, and to make them work together:

1. **A diagnosis of grounding under compression** (Sec. 3). QA-oriented pruners empty frames and fall below random selection; in reasoning-style answers the decoder's video-free prior fills the gaps, which an evidence-removal test shows causally.
2. **Coverage**, a per-layer equal quota filled by the question (Sec. 4.1). Under the standard one-line prompt it matches the uncompressed model with 24% of the visual tokens on Qwen3-VL-8B and the grounding-tuned TimeLens-8B, beats random selection, SemVID and a matched low-

resolution input, and at 897 tokens exceeds a 1 fps input of 1,125 tokens by $+8.8^*$ mIoU; on the full Charades-STA set it matches the uncompressed model and beats random selection (Sec. 6.2).

3. **PCD and an answer-format study** (Secs. 4.2 and 6.2). PCD helps reasoning-style answers on all eight memories we test and on both models; with 10% of the tokens CoCo improves on the uncompressed model by $+11.2^*$ and $+10.7^*$ mIoU on S-EMBER grounded QA under the same prompt. The same study shows that the answer format alone moves single-moment grounding by about 30 mIoU and decides whether contrastive decoding helps.
4. **Floor-adjusted, reproducible evaluation** (Sec. 5). Video-free floors and a length-rescaling baseline for every method, paired bootstrap tests, a reproduction of the published TimeLens-Bench numbers within 0.3 mIoU, and a broad set of negative results, including the limits of Coverage on short, densely sampled videos.

2. Related Work

Frame selection. BOLT [32] scores frames against the query with a contrastive image–text model and selects them by inverse transform sampling; AKS [40], MDP3 [5], FOCUS [2] and Q-Frame [6] optimise relevance together with coverage or diversity. All are evaluated on multiple-choice QA. Our optional retrieval front end (Sec. 4.3) is BOLT adapted to grounding, and we find that it contributes little once tokens are chosen by the question.

Visual token pruning for video LLMs. Query-agnostic methods compress by redundancy or saliency: FastVID [38] segments the video and keeps density peaks, VidCom^2 [33] gives each frame a budget by its uniqueness, FlashVID [17] selects by [CLS] attention and merges tokens across frames with trees, and DyCoke [41] and FrameFusion [20] merge similar tokens. Query-aware methods use the text: SemVID [14] scores tokens by word-embedding relevance with MMR diversity, AdaTP [1] boosts segments similar to the text, LGTTP [4] reads temporal words in the question to reallocate tokens, and PruneVid [25] keeps the top 40% of tokens by question attention at one LLM layer. KTV [12] is closest in structure (keyframes, then tokens with a relevance-weighted per-frame budget), and LGTTP is the only method we know that evaluates grounding, at a milder budget and offline. CoCo differs in three ways: the per-frame quota is equal by design, the score is the actual question's attention propagated exactly through all layers, and the quota exists to keep every timestamp, which we show matters for grounding.

KV-cache compression and video memory. SnapKV [30], H_2O [48], PyramidKV [15] and Ada-KV [19] evict KV entries by attention in long-context LLMs; our token stage is SnapKV-style eviction applied to video with a per-frame constraint. For streaming video, HERMES [10] compresses the cache with generic pseudo-queries, ReKV [16] retrieves KV blocks at question time, and StreamMem [8] and InfiniPot-V [3] maintain bounded memories. These works report QA only. Our memory is built per question from the frames up to the question time: it is causal, but it is not a persistent streaming memory, and we make no claim about the cost of building it (Sec. 7).

Contrastive decoding and prior correction. Contrasting a model's prediction with a weaker or less informed one is an established decoding tool. Contrastive Decoding [29] subtracts a small "amateur" model and introduced the plausibility constraint we use; classifier-free guidance for language models [37] and context-aware decoding [39] amplify the difference between the predictions with and without the conditioning context, which is the form of our Eq. (4). The idea of measuring the prior with a content-free input and correcting for it is older still: contextual calibration [49] rescales label probabilities by the model's answer to "N/A", and domain-conditional PMI [22] reweighs answers by their task-conditioned prior. For vision-language models, VCD [28] contrasts against a noised image, ICD [43] against a disturbed instruction, M3ID [18] against the absence of the image, and PAI [31] amplifies image attention while subtracting text-only logits; SEASON [7] extends the idea to video with temporally corrupted negatives. These methods target hallucination, faithfulness or classification. PCD uses the same contrast for a different purpose: it is applied only to *time values*, with a confidence-adaptive strength, and we study when it helps and when it does not (Secs. 6.2 and 6.5), including whether its benefit is more than a uniform lengthening of the interval (Sec. 6.5).

Grounding-tuned video LLMs. TimeChat [36], VTimeLLM [23] and LITA [24] train video LLMs to localise moments, with timestamp-aware encoders, boundary-aware training stages or dedicated time tokens; TimeLens [47] re-annotates the standard grounding benchmarks and fine-tunes Qwen-VL models with reinforcement learning. These works improve grounding with full visual memories. We use TimeLens-8B as a grounding-tuned model and TimeLens-Bench as the standard protocol, and ask what happens to such a model when its memory is pruned.

Training-free temporal grounding. NumPro [45] draws frame numbers on the frames, FV-Action [9] scans the video with yes/no questions, REZE [13] recasts grounding as recognition, and [Authors to verify] [11] read intervals from calibrated attention heads, debiased against a blank-video run. All assume the full set of video tokens. CoCo addresses the complementary problem of grounding from a *compressed* memory; these readouts could run on top of it **[TODO: run FV-Action and the attention readout on the same 10% memory]**.

Benchmark biases. Otani et al. [34] showed that blind baselines are competitive on Charades-STA and ActivityNet Captions because of positional priors in the annotations. We find the same on the datasets we use, adopt their blind baselines as video-free floors in every comparison, and add a length-rescaling floor for methods that lengthen intervals.

3. Grounding under a Token Budget: a Diagnosis

This section establishes Observations 1, 2 and 4 of the introduction on S-EMBER, before any method is introduced; Observation 3, on the answer format, is studied in Sec. 6.2. Unless stated otherwise, numbers in this section use random 10% pruning as the reference memory.

3.1. Setting

A question x is asked at time q about a video $v_{0:q}$. The model sees F=64 frames from $[0, q]$, encoded into a KV cache $\mathcal{M}$ of N visual tokens interleaved with timestamp tokens (one "`<t seconds>`" per pair of frames in Qwen3-VL). A budget ρ (default ρ=0.1) limits the number of tokens kept in each layer to ρN, with kept timestamps counted inside the budget. The model answers with a short *timeline*:

```
Seen: t_1 seconds, …
Seen: t_k seconds, …
Answer: …
Time: [s, e]
```

from which the interval $[s, e]$ is parsed. This reasoning-style format is natural for grounded QA, where the answer explains what was seen before it states the interval, and the whole diagnosis uses it. Section 6.2 compares it with the one-line format of grounding benchmarks (*"The event happens in s–e seconds"*). Grounding is scored by its IoU with the gold interval (mIoU and R@$\{0.3, 0.5, 0.7\}$); grounded QA additionally requires the answer to be judged correct (GQ@0.5: correct answer and IoU $\geq$ 0.5). We call an answer an *early start* when its interval starts in the first 5% of the window, and a *leak* when it starts early although the gold evidence starts later.

3.2. QA pruners lose temporal coverage

At 10% of the tokens, FastVID, VidCom^2, FlashVID and SemVID all score below random token selection (Sec. 6.3). The cause is visible in where the kept tokens lie (Tab. 1). HERMES's attention-plus-recency score keeps 27.6% of the tokens in the last quarter of the window and only 2.5–4.7% elsewhere; FastVID keeps a token from only 25% of the sampled frames in a median question, because it retains one representative per run of similar frames, and runs of similar frames are exactly where a continuing event lives. Random selection keeps about 9% everywhere and never empties a frame. A timestamp is useful only when visual content of its frame survives with it, so a selector that empties frames leaves holes in the timeline. Holes hurt even when the score itself is reasonable: ranking HERMES's scores within each frame (an equal per-frame share, "stratified") instead of globally gains 5.3* mIoU.

Table 1. **Where pruning keeps tokens.** Share of the visual tokens kept, by position in the window (S-EMBER, 10% budget), and the resulting grounding (mIoU) and leak. FastVID (mIoU 17.9, leak 38%) keeps 6.0 / 8.5 / 10.5 / 10.6 / 12.9% over the five fifths of the window, and tokens from only 25% of the frames.

Selector	0–10%	10–25%	25–50%	50–75%	75–100%	mIoU	Leak
HERMES	4.7	3.9	2.6	2.5	27.6	18.3	49%
Stratified	8.7	8.7	8.6	8.6	8.7	23.7	**29%**
Random	8.9	8.8	8.6	8.7	8.6	**25.8**	30%

3.3. The prior fills the gaps

The model answers without video. Asked every question with an empty visual memory, the model still writes a timeline: 89% of its first `Seen:` moments are at 0 s and 90% of its intervals start in the first 5% of the window (88% leak). Its intervals are 0.45× the gold length (median) and 87% of its ends are more than half an interval too early. We call this the model's *temporal prior*: start early, stop soon.

The prior leaks into answers with video. The same signature appears in every memory, weaker (Fig. 2). About 30% of intervals start early at every random budget and with no pruning at all, against 16% of gold intervals, and the share rises when coverage is lost (HERMES 49%, HERMES streaming 37–39%). The leak is therefore not a matter of the number of tokens, but of where the memory is silent.

Removing the evidence causes the leak. To test causality we drop the uniform frames that fall inside the gold interval ("evidence") or the same number of frames elsewhere ("control"), keeping every other frame with its true timestamp (Tab. 2). With the evidence removed, the answers do not spread out: they move to the start of the video, and the leak rises by 14 points for both the unpruned and the random 10% memory. Within the HERMES 10% memory, questions whose evidence keeps under 5% of its tokens leak 56% of the time, against 33% when at least 15% survives.

An instruction does not remove it. Adding "The relevant moments may occur at any point in the video; do not assume they are near the start" to the prompt leaves the leak almost unchanged (29.5% → 28.0%) and grounding unchanged (−0.1 mIoU). Within this answer format, the prior acts in the output distribution and has to be removed there. Whether the same prior appears when the model answers with a single interval is an open question: there PCD has no effect (Sec. 6.2), which suggests that the prior decides fewer of the written times. **[TODO: no-video answers and leak**

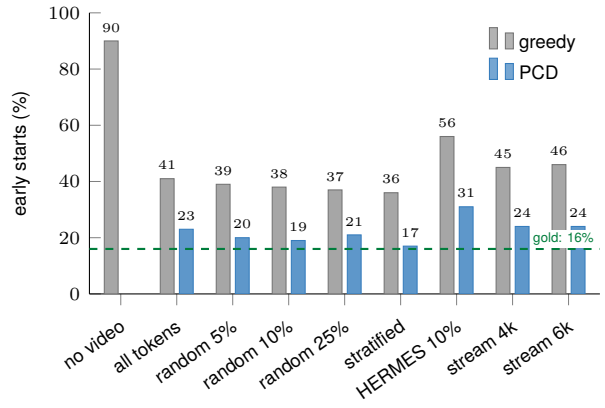


Figure 2. **The temporal prior leaks into every memory, and PCD removes most of it.** Share of predicted intervals that start in the first 5% of the window (S-EMBER). Without video the model starts early 90% of the time; every memory inherits part of this habit, most where pruning loses coverage (HERMES). PCD roughly halves it everywhere.

Table 2. **Causal test.** Frames inside the gold evidence ("evidence") or as many frames elsewhere ("control") are removed. Acc. / mIoU / GQ@0.5 and leak; leak measured on the 397 questions whose evidence starts after the first 5% of the window. PCD: Sec. 4.2; earlier runs: PCD without its answer reserve.

		Greedy		PCD	
Memory	Removed	Acc./mIoU/GQ	Leak	Acc./mIoU/GQ	Leak
Random 10%	control	16.6/28.8/8.8	**31%**	15.4/34.6/10.5	**11%**
	evidence	7.8/ 8.6/1.7	46%	9.3/10.3/1.1	23%
All tokens	control	**19.6/29.4/10.1**	37%	**18.1/35.9/12.6**	15%
	evidence	9.1/ 9.4/1.9	51%	9.1/10.5/1.5	22%

under the one-line and direct-answer prompts.]

The end is the weak boundary. Across memories, the median end error is 0.40–0.69 of the gold length without contrastive decoding, against 0.14–0.30 for the start, and the most frequent failure is an interval entirely *before* the gold one (33% of all answers, Tab. 25).

3.4. Benchmark floors

Positional regularities make some intervals predictable without the video (Tab. 3). On S-EMBER, questions are asked while the stream is running and 67% of gold intervals end within the last 20% of the window, so the constant $[0.15q, q]$ scores 36.9 mIoU. On ActivityNet Captions, whose moments cover 38% of the video on average, $[0, 0.95q]$ scores 37.9; on Charades-STA, $[0, 0.45q]$ scores 29.7. QVHighlights is the least biased (26.8 mIoU, 17.6 R@0.5). These constants are tuned on the test questions, so they upper-bound any video-free rule of this form. On three of the four datasets the best constant beats the uncompressed model answering with a timeline (with a one-line answer, Qwen3-VL-8B is well above it on ActivityNet and Charades-STA, Sec. 6.2), which makes the floors a necessary reference for every comparison that follows.

Table 3. **Video-free floors.** Best constant interval $[aq, bq]$ on a 20×20 grid, tuned on the test questions (each metric uses its own best constant; $[a, b]$ is the mIoU optimum), against the uncompressed model. Closed: share of questions with gold end $< 0.8q$.

	Best constant					All tokens	
Dataset	$[a, b]$	mIoU	R@0.5	R@0.7	Closed	mIoU	R@0.5
S-EMBER	$[.15, 1]$	36.9	34.7	24.2	33%	28.0	26.7
QVHighlights	$[0, 1]$	26.8	17.6	8.8	62%	29.1	28.6
ActivityNet	$[0, .95]$	37.9	35.8	20.8	56%	22.9	18.2
Charades-STA	$[0, .45]$	29.7	29.6	15.4	70%	16.6	14.2

4. Method: CoCo

CoCo acts at the two places where Sec. 3 locates the failures: the memory (Coverage) and the decoder (PCD). Neither needs training. Figure 3 gives an overview.

4.1. Coverage: equal quotas filled by the question

After the F frames are encoded, the memory is pruned once, independently in each of the L layers. Let $\mathcal{T}_f$ be the visual tokens of frame pair $f \in \{1, \dots, F'\}$ and B the per-layer budget for visual tokens.

Question relevance. The question x is appended to the full, uncompressed memory and propagated exactly through all layers, so that the queries at layer ℓ come from the real hidden states. The relevance of visual token i at layer ℓ is the attention it receives from the question, averaged over heads h and question positions j:

$$a_\ell(i) \;=\; \frac{1}{H\,|x|} \sum_{h=1}^{H} \sum_{j \in x} \mathrm{softmax}_i\Big(\tfrac{1}{\sqrt{d}}\, Q_j^{(\ell,h)}\, K_i^{(\ell,h)\top} \Big). \tag{1}$$

Equal quota. Every frame pair receives the same share of the budget, $B_f = \lfloor B/F' \rfloor$ (remainders assigned in order), and layer ℓ keeps

$$\mathcal{K}_\ell \;=\; \bigcup_{f=1}^{F'} \mathrm{top}_{B_f} \big\{\, a_\ell(i) \;:\; i \in \mathcal{T}_f \,\big\}. \tag{2}$$

The timestamp tokens of f are kept if and only if $\mathcal{K}_\ell \cap \mathcal{T}_f \neq \emptyset$. With an equal quota every frame pair keeps at least one token at any practical budget, so *every time anchor survives*, and the question decides which part of each frame is shown. Kept tokens retain their original rotary positions, so their temporal order is preserved.

Why equal, and why the question. The equal quota is a deliberately simple coverage constraint. It guarantees the

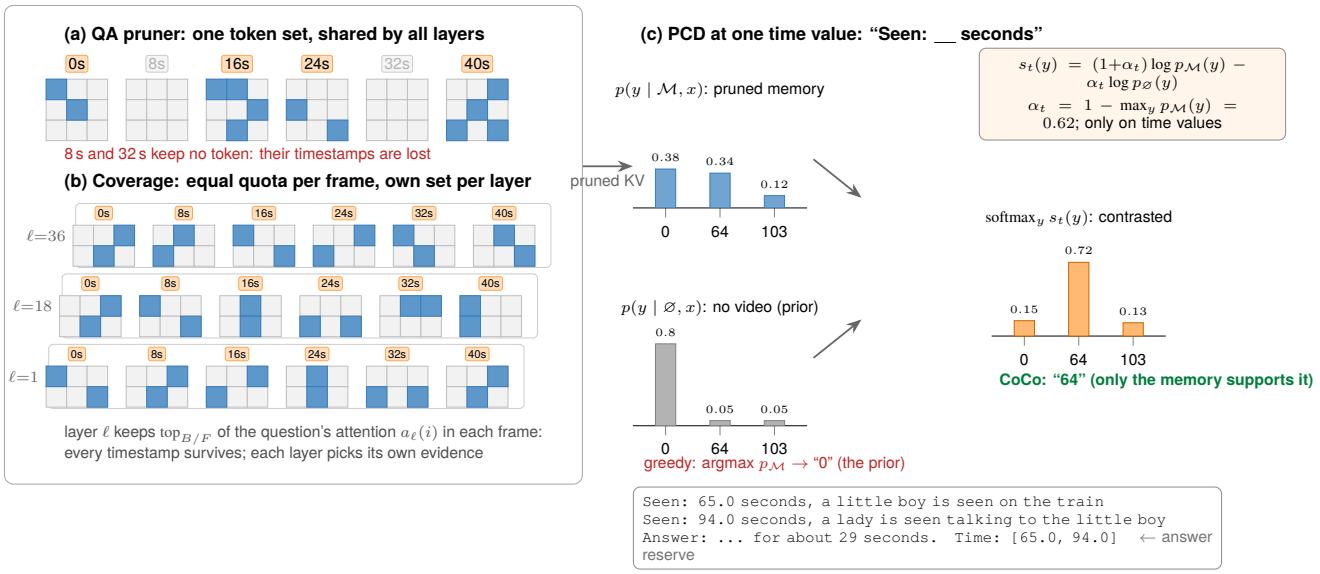


Figure 3. **Overview of CoCo. (a)** QA-oriented pruners keep one set of salient or unique tokens in every layer and can empty whole frames, removing their timestamps. **(b)** Coverage prunes each layer separately: every frame keeps the same share of the layer budget, filled with the tokens that the question attends to most *in that layer*. Every timestamp survives in every layer, and different layers keep different evidence (sharing one set across layers costs 1.5 mIoU, Tab. 15). **(c)** PCD at one time value of the teaser example (probabilities illustrative). Greedy decoding picks "0", which the memory supports only as much as the video-free prior does; the contrasted score discounts it and promotes "64", which only the memory supports. The answer reserve guarantees that the final interval is written (output shown is the real CoCo answer).

timeline, which grounding needs and multiple-choice QA does not, and it stops attention from collapsing onto a few frames: when quotas follow attention mass, attention sinks and recency bias give a single frame up to 227 of about 2,140 kept tokens. Relevance-weighted quotas with a floor, from SigLIP similarity or attention mass, did not beat equal quotas in a pre-registered test (-0.2 mIoU). Using the *actual* question, rather than the generic probe questions of streaming memories such as HERMES [10], is what makes attention informative: generic probes with a global top-k are the weakest variant we measured (18.3 mIoU).

Per-layer selection. Each layer keeps its own set $\mathcal{K}_\ell$. Forcing one shared set (attention averaged over layers) costs CoCo -1.5^* mIoU, whereas it does not change random selection ($+0.7$, n.s.). The benefit of per-layer selection is therefore specific to question-driven scores, and it does not explain why random selection beats input-level pruners.

Frames and budget. An equal quota needs enough tokens per frame to show anything: with 2–4 tokens per frame pair, the top-attended tokens carry little content and Coverage collapses (Sec. 6.7). When the budget is small, we therefore spend it on fewer frames: at a fixed number of kept tokens, 32 frames with twice the share per frame recover what 64 frames lose. As a rule we use F=64 when each frame pair keeps at least $\sim$8 tokens and halve F otherwise. **[TODO: this rule was derived from the results of Sec. 6.7; validate it on held-out data.]**

Cost. Coverage adds one forward pass of the question over the uncompressed memory per question; the full memory is prefilled first, so Coverage reduces the memory the decoder reads, not the cost of building it. PCD decodes a second, prompt-only row in lockstep at every step (its scores are used only inside time values), so it roughly doubles the decoder's per-step work on a much shorter context. **[TODO: measured prefill, time to first token, total latency, peak memory and KV size; the current implementation runs 3–4 scoring passes, reducible to one.]**

4.2. PCD: prior-contrastive decoding of time values

Let $p_{\mathcal{M}}(y) = p(y \mid \mathcal{M}, x, y_{<t})$ be the next-token distribution with the compressed memory and $p_{\varnothing}(y) = p(y \mid \varnothing, x, y_{<t})$ the distribution with the same prompt and generated prefix but *no visual tokens*. Then

$$\log p_{\mathcal{M}}(y) \;=\; \log p_{\varnothing}(y) \;+\; \underbrace{\log \frac{p_{\mathcal{M}}(y)}{p_{\varnothing}(y)}}_{\text{evidence from } \mathcal{M}}, \qquad (3)$$

so when the memory carries no information about y, the evidence term is flat and the video-free prior decides: this is the failure of Sec. 3.3. Prior-contrastive decoding (PCD) ampli-

fies the evidence term,

$$s_t(y) = (1+\alpha_t)\log p_{\mathcal{M}}(y) - \alpha_t \log p_{\varnothing}(y) = \log p_{\varnothing}(y) + (1+\alpha_t)\log \frac{p_{\mathcal{M}}(y)}{p_{\varnothing}(y)}, \quad (4)$$

and picks $y_t = \arg\max_{y\in\mathcal{V}_\beta} s_t(y)$ over the plausible set $\mathcal{V}_\beta = \{y : p_{\mathcal{M}}(y) \geq \beta \max_{y'} p_{\mathcal{M}}(y')\}$ with β=0.1, as in VCD [28], so that tokens the memory makes implausible are never promoted. Three choices adapt this rule to grounding.

No-video counterfactual. The negative memory is empty rather than noised or shuffled. It isolates exactly the prior measured in Sec. 3.3, and it is cheap: its cache holds only the prompt.

Time scope. PCD is applied only while a time value is written: inside a `Seen:` time before its description and inside `Time: [s, e]`. All other tokens, including the answer text, are decoded greedily from $p_{\mathcal{M}}$, so the answer is unaffected where the memory already speaks.

Confidence-adaptive strength. We set $\alpha_t = \alpha_{\max}\big(1 - \max_y p_{\mathcal{M}}(y)\big)$ with $\alpha_{\max}$=1. Where the memory makes the model confident (*e.g.* evidence that truly starts at 0 s), PCD fades and the early answer is kept; where the model hesitates between times, the evidence term decides.

The timeline prompt matters: PCD acts on the listed `Seen:` moments and then on the interval, so the interval is written from contrasted observations. When the answer is a single interval, PCD has little to correct: with the one-line grounding prompt it changes mIoU by -0.1 (n.s., Sec. 6.2), and with the direct-answer prompt by $+0.4$.

Answer reserve. The last part of PCD is a guard on the output format. Contrastive decoding lengthens the list of observed moments: under PCD the model sometimes keeps writing `Seen:` lines until the generation limit T, and never writes the interval. We reserve room for the answer: if R tokens remain (R=64) and no `Answer:` line has started, the decoder writes "`\nAnswer:`" and continues normally. Throughout the paper, PCD includes this reserve. This is a format constraint, not a heuristic on the content: the interval is still chosen by PCD from the memory. It removes every unparseable output (from 25–68 of 500 to 0) and improves every configuration on every dataset (Tab. 20).

4.3. Optional: retrieval-guided frames

When the window is much longer than F frames, which frames are encoded can matter as much as which tokens are kept. As an optional front end we sample frames as in BOLT [32]. Each second of video is embedded once with SigLIP 2 [42], which gives a causal 1 fps index. The relevance z_j of second $j \leq q$ is its z-scored cosine similarity with the question, and the F frames are taken at evenly spaced quantiles of

$$p(j) = \tfrac{1}{2}\cdot\tfrac{1}{q} + \tfrac{1}{2}\,\mathrm{softmax}(z)_j, \quad (5)$$

so that half of the frames stay uniform. Retrieval is skipped when uniform frames would be less than 1 s apart. Once tokens are chosen by the question, retrieval adds only $+0.8$ to $+0.9$ mIoU (not significant on S-EMBER, QVHighlights or ActivityNet), so we treat it as optional; the main-table runs include it.

5. Experimental Setup

Models. Qwen3-VL-8B-Instruct [35], 64 frames sampled uniformly from $[0, q]$ (or by Sec. 4.3) at native resolution, greedy decoding or PCD, at most 384 new tokens, batch size 8. Unless stated otherwise all methods share the timeline prompt. In Sec. 6.2 we add TimeLens-8B [47], a Qwen3-VL-8B model fine-tuned for grounding, and the one-line prompt of TimeLens-Bench. Batch size changes about half of the greedy outputs and up to about two points of a score; comparisons with the uncompressed model, which runs at batch size 1–2, carry this noise.

Datasets. **S-EMBER** grounded QA [44]: 475 questions (191 duration, 184 counting, 100 location) on the first 300 videos, asked mid-stream; the memory contains only the frames before the question. **QVHighlights** [27] val: 500 queries with a single relevant window (a seed-0 sample of the 1,020 single-window queries), videos of about 150 s. **ActivityNet Captions** [26] val_1: 500 queries, one sentence per video, videos of about 115 s. **Charades-STA** [21] test: 500 queries, videos of about 30 s. For the last three, the question is *"When does the following happen in the video: <query>?"* and the window is the whole video. **TimeLens-Bench** [47]: the re-annotated Charades-STA, QVHighlights and ActivityNet sets, evaluated with their one-line prompt (full sets and a 513-query QVHighlights subset). **NExT-GQA** [46] test: 1,005 questions on about 40 s videos, answer and interval asked in two separate queries. Our QVHighlights and ActivityNet copies are low-resolution re-encodes (153 and about 84 visual tokens per frame pair, against 660 on S-EMBER), so we also report budgets in absolute tokens.

Budget. 10% of the visual tokens of the 64-frame memory, timestamps included (S-EMBER: about 2,140 tokens; QVHighlights: about 520). ActivityNet is reported at 30% (400–900 tokens), because at 10% the memory holds only 120–300 tokens (Sec. 6.6).

Baselines. All tokens (no pruning); random tokens; FastVID [38], VidCom^2 [33], FlashVID [17] (vision-side stages) and SemVID [14], re-implemented and checked to keep the same tokens as the authors' code on test inputs; each with and without PCD. **[TODO: SnapKV-style observation-window eviction, PruneVid, global question top-k without quotas, and FV-Action / attention readout on the same memory.]**

Floor-adjusted evaluation. For every dataset we report three video-free or video-light references. (i) The *best*

constant interval $[aq, bq]$ over a 20×20 grid, tuned on the test questions, which upper-bounds any constant rule. (ii) *Length rescaling*: the no-PCD prediction stretched about its start by a factor fitted once on S-EMBER (1.76) and applied unchanged to the other datasets; it tests whether a gain could come from longer intervals alone. (iii) The split into *closed* questions (gold end $< 0.8q$) and the rest. End-snapping on S-EMBER is discussed in Appendix D.

Metrics and statistics. mIoU and R@$\{0.3, 0.5, 0.7\}$; on S-EMBER also answer accuracy and GQ@0.5, judged with the official S-EMBER judge prompt by a local Qwen3.8-27B-FP8 model, all rows of a table in one judging session. Answers that state times as minutes:seconds are converted to seconds for every method alike (Appendix A). We use a paired bootstrap over questions (2,000 resamples) and mark with * differences whose 95% interval excludes 0. **[TODO: video-clustered bootstrap.]**

Protocol. All design choices (scores, quotas, α, scope, reserve) were made on S-EMBER. The configuration was then frozen and run once on QVHighlights, ActivityNet and Charades-STA. **[TODO: held-out S-EMBER videos as a test split.]**

6. Results

We organise the results by question. Section 6.1 compares CoCo with all baselines and floors on four datasets with the timeline answer format, and Sec. 6.2 repeats the key comparisons under the standard one-line prompt, on a grounding-tuned model and on NExT-GQA. Section 6.3 shows how published pruners behave on grounding. Sections 6.4 and 6.5 analyse the two components. Sections 6.6 and 6.7 study the budget and how to spend it. Section 6.8 compares with upper bounds that use the gold interval, Sec. 6.9 analyses the optional retrieval stage, Sec. 6.10 lists what did not work, and Sec. 6.11 analyses the remaining errors.

Reading the tables. Numbers are mIoU, or mIoU / R@0.5 where two are given. Δ is a paired difference; * marks differences whose 95% paired bootstrap interval excludes 0. Tables built from the early S-EMBER phases (Tabs. 16 to 19, 23 and 25 and Tab. 2) predate the answer reserve, so PCD there runs without it, and use the official parser and batch size 1; on S-EMBER the two parsers differ by at most 0.7 mIoU and batch size changes no paired result. In every table the best value is in **bold** and the second best underlined, per column unless a caption says per row; floors and differences (Δ) are not ranked.

6.1. Main results

Tables 4 and 5 lead to three conclusions.

(i) With the timeline prompt, CoCo nearly doubles the uncompressed model on the least biased benchmark. On QVHighlights, with 10% of the tokens, CoCo reaches 54.8 R@0.5 (R@0.7: 30.2) against 28.6 (18.4) for all tokens and 25.0 (16.2) for random tokens, and clears every floor by a wide margin (+22.1 mIoU over the best constant, +8.6 over length rescaling). This comparison holds the prompt fixed; the one-line prompt of TimeLens-Bench gives the uncompressed model far higher scores, and there the gain of CoCo is smaller (Sec. 6.2).

(ii) On S-EMBER, the gain is real where the window end is not a shortcut. CoCo beats all tokens, random selection and SemVID under the same decoding. Against the floors its pooled gain is small and not significant, because two thirds of S-EMBER intervals end at the question time. On closed intervals, where that shortcut does not exist, CoCo gains $+7.6^*$ mIoU over the best constant and $+7.5^*$ over length rescaling (Tab. 26).

(iii) On ActivityNet and Charades-STA, the timeline prompt is the bottleneck. With this prompt and our low-resolution copies, the uncompressed model is far below the constant floor on both (22.9 *vs*. 37.9 and 16.6 *vs*. 29.7); with the one-line prompt of TimeLens-Bench it scores 47.1 and 48.5 mIoU (Sec. 6.2). CoCo improves on all tokens and on every pruner on ActivityNet, but no method reaches the floor, and on Charades-STA Coverage is worse than random selection (Sec. 6.7). We report these as limits, not successes.

Answers. CoCo also improves the answers (Tab. 6): accuracy 21.7 *vs*. 17.3 for random tokens ($+4.4^*$) and 19.8 for all tokens, and GQ@0.5 13.3 *vs*. 7.4 and 8.8. Coverage alone gives the best accuracy (22.5); PCD trades about one point of accuracy for its grounding gain, and differences below about two points of accuracy are within the decoding noise of the judge setup. **[TODO: SemVID + PCD (with reserve) in the same judging session.]**

Question types. CoCo improves duration and counting questions most (Tab. 7). Location questions ("where did I put …") remain essentially unsolved by every method (mIoU $\leq$ 11): 62 of the 100 location questions have annotators who disagree on the interval (mean pairwise IoU < 0.6), which caps any method.

6.2. Standard prompt, a grounding-tuned model, and short videos

The results above use one model and the reasoning-style timeline prompt on every dataset. We now test the method under the standard one-line grounding prompt of TimeLens-Bench [47], on a grounding-tuned model (TimeLens-8B [47]), and on NExT-GQA [46]. Unless stated otherwise, paired differences use the same questions; unpruned runs use batch size 1–2 and pruned runs batch size 8, which alone moves scores by up to about two points (Sec. 5).

Our pipeline reproduces published numbers. With the TimeLens evaluation script, Qwen3-VL-8B scores 48.5,

Table 4. **Grounding with 10% of the visual tokens** (ActivityNet: 30%), mIoU / R@0.5. Grey rows are floors: the best video-free constant interval, tuned on the test questions, and the Retrieve + Coverage prediction without PCD lengthened by a factor fitted on S-EMBER. Coverage rows include optional retrieval, which the coverage gate disables on Charades-STA. Empty cells: not yet computed. S-EMBER closed intervals: Tab. 26.

Method	S-EMBER	QVHighlights	ActivityNet (30%)	Charades-STA
Best constant interval	*36.9 / 34.7*	*26.8 / 17.6*	*37.9 / 35.8*	*29.7 / 29.6*
Length rescaling of $R + C$	*38.3 / 40.0*	*40.2 / 42.4*	*34.0 / 33.0*	*24.0 / 17.6*
All tokens (no pruning)	28.0 / 26.7	29.1 / 28.6	22.9 / 18.2	16.6 / 14.2
Random tokens	25.6 / 24.6	26.9 / 25.0	22.9 / 18.6	17.4 / 13.8
FastVID [38]	17.9 / 15.8	19.2 / 15.4		13.3 / 11.6
VidCom^2 [33]	20.4 / 18.7	19.9 / 18.2		12.3 / 8.4
FlashVID [17]	22.7 / 21.5	22.8 / 20.4	20.5 / 15.4†	14.1 / 10.8
SemVID [14]	23.3 / 20.6	24.5 / 21.2	22.8 / 18.6	14.6 / 11.2
Random + PCD	35.0 / 33.9	37.8 / 39.2	31.4 / 29.4	**26.2 / 24.0**
SemVID + PCD	34.8 / 35.8	34.8 / 34.8	30.2 / 27.8	21.7 / 19.2
Coverage only	32.5 / 34.1	34.5 / 35.6	25.4 / 21.2	16.4 / 13.6
CoCo (Coverage + PCD)	**39.3 / 40.8**	**48.8 / 54.8**	**34.7 / 32.4**	23.3 / 21.2

† at 10%.

Table 5. **Paired gains of CoCo in mIoU** (*: 95% paired bootstrap interval excludes 0).

CoCo *vs.*	S-EMBER	QVHighlights	ActivityNet (30%)	Charades-STA
All tokens	$+11.2^*$	$+19.7^*$	$+11.8^*$	
Random + PCD	$+4.2^*$	$+11.0^*$	$+3.3^*$	-2.9^*
SemVID + PCD	$+4.5^*$	$+14.1^*$	$+4.5^*$	
Best constant	$+2.4$	$+22.1^*$	-3.1^*	-6.4^*
Length rescaling	$+1.0$	$+8.6^*$	$+0.8$	-0.7

Table 6. **Grounded QA on S-EMBER** (10%; one judging session). ΔAcc. *vs.* random tokens.

Method	Acc.	GQ@0.5	mIoU	ΔAcc.
All tokens	19.8	8.8	28.0	+2.5
Random tokens	17.3	7.4	25.6	–
FastVID	15.2	4.8	17.9	−2.1
VidCom^2	16.2	5.9	20.4	−1.1
FlashVID	17.3	6.1	22.7	+0.0
SemVID	19.8	7.6	23.3	+2.5
Coverage, uniform frames	**22.5**	11.2	31.7	$+5.3^*$
Random + PCD	20.4	12.2	35.0	+3.2
SemVID + PCD$^\circ$	17.9	9.9	32.3	+0.6
CoCo	21.7	13.3	**39.3**	$+4.4^*$
CoCo + coverage gate	22.1	**13.5**	**39.3**	$+4.8^*$

$^\circ$ earlier runs: PCD without its answer reserve.

Table 7. **S-EMBER by question type**, mIoU / R@0.5 (10%).

Method	Duration (191)	Counting (184)	Location (100)
All tokens	37.6 / 40.3	30.1 / 25.5	5.9 / 3.0
Random tokens	33.2 / 35.1	29.6 / 26.1	3.5 / 2.0
Coverage	43.2 / 46.6	35.3 / 36.4	7.1 / 6.0
Random + PCD	41.0 / 40.8	41.9 / 40.2	**11.0 / 9.0**
SemVID + PCD	38.9 / 36.6	44.6 / **50.0**	8.9 / 8.0
CoCo	**47.6 / 50.8**	**46.9** / 48.4	9.3 / 8.0

59.6 and 47.1 mIoU on the full Charades-STA, QVHighlights and ActivityNet sets of TimeLens-Bench (published: 48.3, 59.4, 46.8), and TimeLens-8B 55.4, 65.7 and 53.2 (published: 55.2, 65.5, 53.2). On the same queries, our own pipeline without pruning matches the script: +0.5 (Charades, 400 queries), +1.4 (ActivityNet, 400) and 0.0 (QVHighlights, 513) mIoU, none significant (Tab. 27).

The timeline prompt is costly for single-moment grounding. Table 8 changes only the prompt. On QVHighlights queries, asking for a timeline of observations before the interval costs about 30 mIoU against the one-line prompt (-30.0^* without pruning, -31.0^* for Coverage). PCD recovers part of the loss ($+13.2^*$ on Coverage, $+10.4^*$ on random tokens) but stays far below the one-line prompt (-17.8^*), and under the one-line prompt it has no effect (-0.1, n.s.; also -0.3 and -0.5 with retrieval). With the timeline prompt, Coverage is not better than random selection on these queries (-2.8^* greedy, $+0.1$ with PCD). Two

Table 8. **Answer format** (QVHighlights, 513 TimeLens-Bench queries, Qwen3-VL-8B, 300 frames, 31% of the tokens), mIoU. One-line: the TimeLens-Bench prompt. Timeline: the prompt of all other tables. PCD°: without the answer reserve (the only one-line PCD run).

	One-line prompt		Timeline prompt	
Selection	greedy	PCD°	greedy	PCD
All tokens	60.5	–	30.4	–
Random tokens	58.2	–	**33.4**	43.9
Coverage	**61.7**	61.6	30.7	43.9

Table 9. **Standard grounding prompt** (QVHighlights, 513 TimeLens-Bench queries, one-line prompt), mIoU / R@0.5. Top: 300 frames at protocol size; bottom: 64 frames chosen by retrieval. Kept: visual tokens in memory. TimeLens input: the TimeLens-Bench input re-sampled to the same number of tokens. Keep ratios count timestamp tokens too: 31% of all tokens is 24% of the visual tokens.

Method	Kept	Qwen3-VL-8B	TimeLens-8B
All tokens	13,650	60.5 / 64.9	67.1 / 73.1
TimeLens input, 2 fps low-res	4,200	58.0 / 62.0	64.3 / 68.8
Random tokens (31%)	3,271	58.2 / 60.8	65.3 / 70.4
SemVID (31%)	4,232	58.8 / 62.4	–
Coverage (31%)	3,271	**61.7 / 66.7**	**67.3 / 73.9**
TimeLens input, 1 fps low-res	1,125	49.5 / 53.0	57.5 / 61.8
Retrieve + random (23%)	897	55.5 / 60.8	–
Retrieve + Coverage (23%)	897	**58.3 / 63.7**	**64.3 / 71.2**

consequences follow. First, the QVHighlights, ActivityNet and Charades-STA numbers of Tab. 4 are relative to a format that handicaps the uncompressed model; the gain over "all tokens" there (Sec. 6.1) does not hold against the one-line prompt. Second, PCD corrects time values in reasoning-style answers, where the model writes several times before the interval; when the answer is a single interval, there is little for it to correct. **[TODO: one-line PCD with the reserve; timeline PCD without it; the same study on TimeLens-8B.]**

Under the standard prompt, Coverage transfers to both models. With the one-line prompt (Tab. 9), Coverage at 24% of the visual tokens beats random selection on both models (+3.5* and +2.0* mIoU), beats SemVID (+2.9*), and beats the TimeLens input re-sampled to a similar budget (+3.7*, +3.0*). It matches the uncompressed model (+1.2 for Qwen3-VL-8B, with a lower bound just at zero; +0.2 for TimeLens-8B). At 897 tokens, retrieval with Coverage exceeds a 1 fps TimeLens input of 1,125 tokens by +8.8* and +6.8*, and retrieval with random tokens by +2.8*. At this budget the grounding-tuned model does lose accuracy: TimeLens-8B with retrieval and Coverage is 2.8* mIoU below its uncompressed memory, whereas at 3,271 tokens it is level (+0.2).

Table 10. **Full TimeLens-Bench sets** (Qwen3-VL-8B, one-line prompt, TimeLens input at 2 fps, 31% of the tokens), R@0.5 / mIoU. Unpruned: the TimeLens evaluation script. QVHighlights: the 1,030 queries finished so far for both methods (random selection still running). Paired ΔmIoU below.

Method	Charades-STA (3,363)	QVHighlights (1,030)
All tokens (no pruning)	**54.0** / 48.5	**65.4 / 60.6**
Random tokens (31%)	52.3 / 48.2	63.2 / 58.9
Coverage (31%)	**54.0 / 48.6**	65.2 / 60.5
Coverage *vs.* random	+0.5*	+1.6*
Coverage *vs.* all tokens	+0.1	−0.1

Table 11. **A grounding-tuned model on S-EMBER** (10%, timeline prompt), mIoU / R@0.5. Answers of TimeLens-8B not yet judged.

Method	Qwen3-VL-8B	TimeLens-8B
All tokens (no pruning)	28.0 / 26.7	33.1 / 34.1
Random tokens	25.6 / 24.6	28.2 / 26.7
Random + PCD	35.0 / 33.9	38.4 / 39.4
Retrieve + Coverage	32.5 / 34.1	37.2 / 37.9
CoCo	**39.3 / 40.8**	**43.9 / 46.9**
CoCo *vs.* all tokens	+11.2*	+10.7*
CoCo *vs.* Random + PCD	+4.2*	+5.5*

The same holds on the full sets. On the complete Charades-STA set of TimeLens-Bench (Tab. 10), Coverage at 31% of the tokens matches the uncompressed model (+0.1 mIoU, n.s.; R@0.5 54.0 *vs.* 54.0) and beats random selection (+0.5* mIoU, +1.6* R@0.5, +1.4* R@0.3), whose R@0.5 and R@0.3 fall significantly below the uncompressed model (−1.7*, −1.0*). On the 1,030 QVHighlights queries finished for both methods, Coverage beats random selection on every metric (+1.6* mIoU, +2.0* R@0.5, +1.7* R@0.7) and is level with the uncompressed model (−0.1), while random selection is below it (−1.7*). Charades-STA videos are short, so 31% leaves about 3,350 of 10,800 visual tokens and the gap to random is small, as on NExT-GQA below. **[TODO: rest of the sweep: QVHighlights random (2 chunks), ActivityNet, all TimeLens-8B cells, and the matched-budget TimeLens inputs.]**

On grounded QA, the full method transfers to a grounding-tuned model. S-EMBER answers explain before they ground, so the timeline format is natural there. TimeLens-8B, trained for grounding, is 2.6–5.1 mIoU above Qwen3-VL-8B in every row of Tab. 11, and the ordering of the methods is the same for both models. CoCo improves on the uncompressed TimeLens-8B by +10.7* mIoU (46.9 *vs.* 34.1 R@0.5), on random selection by +15.7*, and on Random + PCD by +5.5*. Each component helps on its

Table 12. **NExT-GQA** (test, 1,005 questions, 64 frames, two-query protocol: one query for the answer, one for the interval; no PCD).

Model	Method	Acc@QA	Acc@GQA	mIoP	mIoU	IoU@0.5
Qwen3-VL-8B	All tokens	79.7	38.9	**48.0**	**37.7**	**38.5**
	Random 30%	78.6	38.6	47.5	37.3	37.2
	SemVID 30%	78.9	38.5	46.2	35.8	35.3
	Coverage 30%	**79.8**	**39.4**	47.9	37.6	37.8
	Random 10%	75.3	34.9	45.2	35.3	35.1
	Coverage 10%	78.4	36.1	44.6	34.8	34.6
TimeLens-8B	All tokens	**76.7**	**41.5**	**52.1**	**39.8**	**39.5**
	Random 10%	72.3	36.9	48.9	37.7	37.9
	Coverage 10%	75.4	38.5	49.3	37.8	37.6

own: PCD adds $+10.2^*$ to random tokens and $+6.7^*$ to Retrieve + Coverage; Retrieve + Coverage adds $+9.0^*$ to random tokens. **[TODO: judge TimeLens-8B answers (Acc, GQ@0.5).]**

On short videos, Coverage keeps the answers but not more of the interval. NExT-GQA videos last about 40 s, so 64 frames already sample them densely (Tab. 12). At 10% of the tokens, Coverage answers better than random selection on both models ($+3.1^*$ Acc@QA each), but grounds no better (mIoU -0.5 and $+0.1$, n.s.), and both lose about three points of Acc@GQA against the uncompressed model (-2.8^*, -3.0^*). At 30%, Coverage is lossless ($+0.5$ Acc@GQA, -0.2 mIoU, n.s.) and grounds better than SemVID ($+1.8^*$ mIoU). This matches Charades-STA (Sec. 6.7): when frames are dense, where tokens are kept matters less than how many. We use the two-query protocol; asking for answer and interval in one reply, as in prior zero-shot reports, lowers Acc@GQA by 16.5^* (Qwen3-VL-8B) and 9.2^* (TimeLens-8B), so absolute numbers are comparable only within a protocol. **[TODO: PCD on NExT-GQA; 5% budget.]**

6.3. QA pruners on grounding

Every published pruner (Fig. 4) falls below random token selection on S-EMBER (-2.3 to -7.6 mIoU, all significant), QVHighlights (-2.4 to -7.7; SemVID not significant) and Charades-STA (-2.8 to -5.2, all significant); on ActivityNet at 10% the input-level pruners are above random, which we discuss in Sec. 6.7. Random selection varies little across seeds (26.0 ± 0.6 mIoU over three seeds on S-EMBER), so the gaps are not noise. The ordering follows coverage (Tab. 1): the methods that empty frames most aggressively lose most, and those that spread tokens lose least. Accuracy, in contrast, is barely affected (Tab. 6): FastVID's accuracy is within noise of random selection (-2.1, n.s.) while its mIoU is 7.6 lower. A pruner built to keep answers keeps answers, not time. PCD helps every pruner ($+7$ to $+12$ mIoU) but does not close the gap to CoCo.

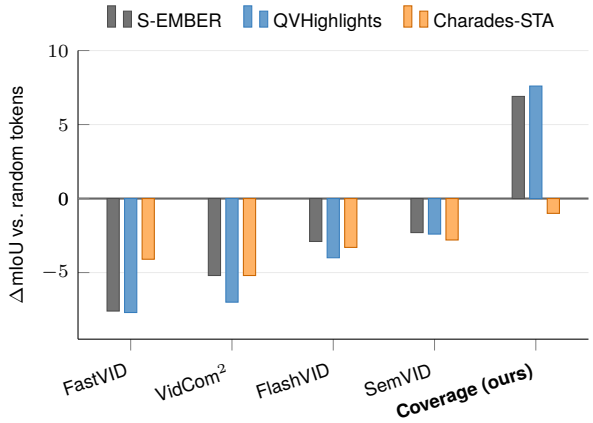


Figure 4. **QA pruners fall below random selection on grounding.** Paired mIoU difference to random token selection at the same 10% budget, greedy decoding. All four published pruners lose on all three datasets (all significant except SemVID on QVHighlights). Coverage gains on the two long-video datasets and is level with random selection on 30 s Charades videos.

Table 13. **Component ladder** (mIoU / R@0.5; ActivityNet at 30%). R: retrieval frames; C: Coverage; PCD: prior-contrastive decoding.

Configuration	S-EMBER	QVHighlights	ActivityNet
Uniform, random tokens	25.6 / 24.6	26.9 / 25.0	22.9 / 18.6
R, random tokens	29.4 / 29.5	28.7 / 28.0	–
C (uniform frames)	31.7 / 32.4	33.7 / 33.0	24.6 / 21.6
R + C	32.5 / 34.1	34.5 / 35.6	25.4 / 21.2
R + C + PCD (CoCo)	**39.3 / 40.8**	**48.8 / 54.8**	**34.7 / 32.4**

6.4. Analysis of Coverage

Contribution. Table 13 adds the components one at a time. Coverage is the largest gain on the memory side: $+6.1^*$ mIoU on S-EMBER and $+6.8^*$ on QVHighlights over random tokens. Retrieval adds $+0.8$ to $+0.9$ on top (not significant on any dataset); once tokens are chosen by the question, which frames are encoded matters less. **[TODO: Coverage + PCD without retrieval on all datasets.]**

Design choices. Table 14 varies one choice at a time. *The quota should be equal*: five relevance-weighted variants, from SigLIP similarity, attention mass or both, with floors of 25–50%, are within ± 0.5 mIoU of the equal quota and none is significant (the uniform-frame SigLIP variant was our pre-registered primary test). Quotas that follow raw attention mass are pulled by attention sinks and recency bias: a single frame received up to 227 of about 2,140 kept tokens. *The score should be the actual question*: HERMES's generic probe questions give 23.7 even with an equal quota, and 18.3 without. *Deterministic top-k beats sampling* from a coverage-floored density (31.7 *vs.* 30.0); with the quota

Table 14. **Design choices of Coverage** (S-EMBER, greedy decoding). Frames: R = retrieval, U = uniform. Δ *vs.* the equal quota with question top-k on the same frames.

Token selection	Frames	mIoU	Δ
Equal quota, question top-k	R	32.5	–
Quota ∝ SigLIP relevance (floor 50%)	R	**33.0**	+0.5
Quota ∝ SigLIP (floor 25%)	R	32.6	+0.1
Quota ∝ SigLIP + attention	R	32.9	+0.4
Quota ∝ attention mass	R	32.1	−0.4
Contrastive token score‡	R	**33.0**	+0.5
One token set shared by all layers	R	31.9	−0.6
Equal quota, question top-k	U	31.7	–
Quota ∝ SigLIP relevance	U	31.4	−0.2
Equal quota, sampling instead of top-k	U	30.0	−1.7
Contrastive token score‡	U	30.5	−1.2
HERMES score, equal quota (stratified)	U	23.7	
HERMES score, global top-k	U	18.3	
Question, global top-k, no quota	U	**[TODO:]**	

‡ $\log a(\text{question}) - \log a(\text{generic sentence})$.

Table 15. **Per-layer *vs.* shared token sets** (S-EMBER, 10%). Δ = shared − per layer.

Method	Per layer	Shared	ΔmIoU
Random tokens	25.6	26.3	+0.7
Random + PCD	35.0	35.8	+0.7
R + C	32.5	31.9	−0.6
CoCo	**39.3**	**37.8**	−1.5*

in place, coverage is already guaranteed. A contrastive token score that subtracts the attention of a generic sentence, meant to cancel sinks, does not help.

Per-layer selection. Table 15 compares one token set per layer with a single set shared by all layers. Sharing does not change random selection, which rules out per-layer diversity as the reason random selection beats input-level pruners. It costs CoCo −1.5* mIoU: different layers attend to different tokens, and question-driven selection benefits from letting each choose.

6.5. Analysis of PCD

In reasoning-style answers, PCD is a plug-in for any memory. Table 16 applies PCD to eight memories: three random budgets, the uncompressed model, HERMES and stratified pruning, and two HERMES streaming memories. It improves mIoU significantly on all eight (+4.6 to +8.0) and GQ@0.5 on six, roughly halves the early starts everywhere, and its gains are largest where the memory is smallest. With 10% of the tokens and random selection it already beats the uncompressed model (32.6 *vs.* 28.0). PCD also transfers to FastVID (+7.6* mIoU, leak 38% → 12%), the pruner that loses coverage most.

The no-video counterfactual is the right one. Table 17 changes only what the memory is contrasted against. The no-video input isolates the prior and removes it (early starts 38% → 19%). Noised frames, as in VCD, still show part of the video *and* still produce the prior, so the two runs disagree mainly about content and little of the prior is removed (+0.5 mIoU). Permuted timestamps leave the prior in place as well. A one-sided rule that only penalises tokens the video argues against is worse than PCD (−5.6* and −4.8*): without video the model is about 99% sure of "0", so even a well-supported "0" looks like evidence against. Boosting visual attention during decoding, or stating the bias in the instruction, leaves the prior in place.

Time scope and adaptive strength. Contrasting every token also changes words that have nothing to do with time: the answer text changed in 85% of questions, lists of more than eight moments rose from 12 to 56, and over-counting rose from 14% to 34% of counting questions. Restricting PCD to time values keeps the grounding gain and disturbs the text less (Tab. 18, top). PCD also risks hurting questions whose evidence really starts at 0 s. With a constant strength, their accuracy on the uncompressed memory falls from 21.8 to 16.7; with the adaptive strength, which fades when the model is confident, it rises to 23.1 (Tab. 18, bottom; +6.4* *vs.* constant). The adaptive rule also gives the best *discrimination*, early starts on early-evidence questions minus early starts on the others (56.4 *vs.* 49.6 for greedy, all-tokens memory).

PCD is near the limit of decoding. We logged both distributions at the first time value of every answer and replayed other decision rules offline (Tab. 19). Even a threshold tuned on the gold labels cannot beat PCD's trade-off between keeping true early starts and removing leaked ones. The remaining early-start errors are cases where the memory itself is unsure; they need evidence, not a better decoding rule.

Is PCD more than a longer interval? PCD lengthens predicted intervals (median predicted/gold length 0.57 → 0.67 on S-EMBER), and longer intervals alone raise mIoU on benchmarks with long gold intervals. The length-rescaling floor tests this directly (Tab. 5, last row). On QVHighlights (+8.6*) and on S-EMBER closed intervals (+7.5*, Tab. 26), CoCo clearly beats rescaling: PCD moves the interval toward the evidence, not just outward. On pooled S-EMBER, ActivityNet and Charades-STA the two are statistically tied, so on those sets the benefit of PCD cannot be separated from a calibrated lengthening. **[TODO: start/end error decomposition on closed questions; mismatched-video and shuffled-frame negatives.]**

Answer reserve. PCD rewards fine time steps that the prior finds unlikely, so it sometimes lists one continuing activity at one-second steps until the token limit. The reserve re-

Table 16. **PCD on eight memories** (S-EMBER, Acc. / mIoU / GQ@0.5; earlier runs: PCD without its answer reserve). Plain PCD: on all tokens with constant α=0.5; PCD: confidence-adaptive strength on time values only (ours). Δ PCD *vs.* greedy. Best and second best per row.

Memory	Greedy	Plain PCD	PCD	ΔmIoU	ΔGQ@0.5	Early starts
Random 5%	15.8 / 23.6 / 5.9	15.4 / 29.2 / 9.1	**16.4 / 31.5 / 9.9**	+8.0*	+4.0*	39% → 20%
Random 10%	15.2 / 25.8 / 7.2	**16.2** / 32.1 / **10.1**	**16.2 / 32.6 / 10.1**	+6.8*	+2.9	38% → 19%
Random 25%	17.7 / 28.2 / 8.2	**18.9** / 33.0 / 11.8	**18.9 / 34.9 / 13.3**	+6.7*	+5.1*	37% → 21%
All tokens	18.5 / 28.0 / 8.6	19.8 / 31.5 / 11.2	**21.5 / 32.7 / 13.3**	+4.6*	+4.6*	41% → 23%
HERMES 10%	13.1 / 18.3 / 4.0	**15.4** / 22.6 / **7.2**	13.3 / **24.1** / 5.9	+5.7*	+1.9	56% → 31%
Stratified 10%	14.1 / 23.7 / 6.1	14.5 / 29.9 / 8.8	**15.8 / 31.0 / 10.5**	+7.3*	+4.4*	36% → 17%
HERMES streaming 4k	13.5 / 21.6 / 4.0	**17.1** / 27.4 / **8.2**	15.4 / **28.6 / 8.2**	+6.9*	+4.2*	45% → 24%
HERMES streaming 6k	14.3 / 22.6 / 4.8	14.7 / 28.1 / 7.6	**15.4 / 29.8 / 8.4**	+7.2*	+3.6*	46% → 24%
Mean of 8	15.3 / 24.0 / 6.1	16.5 / 29.2 / 9.3	**16.6 / 30.7 / 10.0**	+6.7	+3.9	

Table 17. **What to contrast against** (S-EMBER, random 10%). Only the counterfactual input changes; the main input is always the real memory.

Counterfactual	mIoU	GQ@0.5	Early
None (greedy)	25.8	7.2	38%
No video (ours)	**32.1** (+6.2*)	**10.1** (+2.9*)	**19%**
Noised frames (VCD [28])	26.3 (+0.5)	7.8 (+0.6)	30%
Evidence-against rule, α=1 / 2	26.5 / 27.3	–	–
Attention gain (PAI-style)	+1.3 (n.s.)	+0.8	42%
Instruction "do not assume early"	25.7	6.9	37%

Table 18. **Time scope and adaptive strength** (S-EMBER, Acc. / mIoU / GQ@0.5). Top: PCD on all tokens *vs.* time values only. Bottom: the 78 questions whose evidence truly starts in the first 5% of the window. Top: best per row.

Memory	All tokens	Time values only
Random 10%	**16.2** / 32.1 / 10.1	16.0 / 32.1 / **10.3**
Random 25%	**18.9** / 33.0 / 11.8	17.7 / **34.0** / 11.8
All tokens	19.8 / 31.5 / 11.2	**20.8 / 32.9 / 12.4**
HERMES 10%	**15.4** / 22.6 / 7.2	14.7 / **24.3** / **7.4**
Streaming 4k	**17.1** / 27.4 / 8.2	16.2 / **29.2** / **9.3**
Early-evidence questions (78), all tokens memory		
Greedy	21.8 / 49.2 / 16.7	
PCD, constant α	16.7 / 55.3 / 12.8	
PCD, adaptive α	**23.1 / 56.9 / 19.2**	

Table 19. **Decision rules replayed at the first time value** (S-EMBER, random 10%). Share of answers starting at "0".

Rule	Early evidence ↑	Other (leak) ↓
Greedy	**69%**	16%
PCD, α=0.5	46%	**6%**
PCD, adaptive	51%	7%
Evidence-against, α=4	51%	9%
Skewed or tempered prior	50–55%	8–11%
Best threshold (uses gold)	46%	**6%**

Table 20. **The answer reserve inside PCD**: PCD *vs.* PCD without the reserve (paired ΔmIoU), and unparseable answers without → with it.

Configuration	S-EMBER	QVHighlights	Charades
Random + PCD	+1.9*	+1.5*	+2.7*
SemVID + PCD	+2.5*	–	+5.6*
R + C + PCD	+1.8*	+6.1*	+2.4*
Unparseable (of 500)	26–39 → 0	25–60 → 0	28–68 → 0

moves every such failure and improves every PCD configuration we tested (Tab. 20), most where run-ons were frequent (SemVID + PCD on Charades, CoCo on QVHighlights). It costs nothing when the model answers within the limit, which every greedy run does.

Prompt format. PCD needs a reasoning-style answer: with the official direct-answer prompt it does not help (+0.4 mIoU). An *event* timeline that lists start and end per occurrence (`Event: 12.0 -- 15.5 seconds`) improves the greedy baseline by +7.4* mIoU by stating extents, and PCD still adds +2.1 to +3.3 on top (35.0 mIoU, 12.6 GQ@0.5 at 768 tokens). The two fixes overlap: the format corrects extents, PCD corrects the leak. CoCo uses the simpler timeline in all main results. **[TODO: event timeline with CoCo.]**

When does PCD help? The results so far point to one rule: PCD helps in proportion to how much the video-free prior decides the time that is written. Its gain is largest where the memory says least (random 5%: +8.0* mIoU; uncompressed: +4.6*, Tab. 16); the adaptive strength, which fades when the memory makes the model confident, protects true early starts that a constant strength removes (Tab. 18); and with a one-line answer, where the model states a single interval instead of listing observations, PCD has no effect at

all (Sec. 6.2). We state this as a hypothesis that the per-token log-probabilities we already store can test directly. **[TODO: PCD gain against the divergence between $p_{\mathcal{M}}$ and $p_{\varnothing}$ at each written time value, across all conditions; one prediction stated before the run.]**

6.6. Budget

Table 21 and Fig. 5 vary the budget at 64 frames and show two regimes. Above roughly 500 kept tokens, question-driven selection is flat or even improves as the budget *shrinks*: on S-EMBER, Retrieve + Coverage rises from 31.0 (20%) to 34.6 (5%) and CoCo from 36.8 to 39.3, while random selection and SemVID degrade. Pruning by the question removes distractors as well as cost: at 5% CoCo leads SemVID + PCD by $+7.4^*$ mIoU, at 20% by $+1.5$ (n.s.). Below roughly 300 kept tokens (QVHighlights 5%, ActivityNet 10%), the equal quota leaves only 2–4 tokens per frame pair, the top-attended tokens carry little content, and CoCo collapses (9.7 and 11.6), below random selection and below SemVID, whose cross-frame merging degrades more gracefully. The threshold is absolute, not relative: it depends on the number of tokens per frame, which is low in our re-encoded QVHighlights and ActivityNet copies.

6.7. Fewer frames or fewer tokens per frame?

The collapse at small budgets is not a limit of the equal quota itself but of how the budget is spent. In Tab. 22 every row of a block keeps the same number of tokens and only the split between frames and tokens per frame changes. With 32 frames instead of 64, each frame pair keeps twice as many tokens, and CoCo recovers from 9.7 to 43.1 mIoU on QVHighlights and from 11.6 to 33.9 on ActivityNet, where it now beats SemVID + PCD ($+5.5$) and nearly matches its own 64-frame result at three times the budget (34.7). Halving again to 16 frames loses a little: past a point, temporal coverage becomes the bottleneck. The baselines gain from fewer frames too, but less. On Charades-STA (30 s videos) fewer frames do not help any method. We therefore recommend the rule of Sec. 4.1: keep at least about eight tokens per frame pair and reduce F when the budget does not allow it. At equal tokens, more frames with fewer tokens each does not help either: 128 frames at 5% on S-EMBER give $+1.2$ mIoU (n.s.) without PCD and -3.6^* with it, with 74–78 unparseable answers.

6.8. Comparison with gold-informed upper bounds

Table 23 gives the memory information it should not have. At the same budget, PCD without any gold information beats perfect token selection on grounding ($+4.7^*$ mIoU over the oracle selector at 10%, $+4.5^*$ at 5%): knowing which tokens to keep does not stop the prior, and the oracle selector still leaks 33% of the time. Against the strongest oracle, which sees only the evidence, PCD closes most of the grounding gap (GQ@0.5 13.3 *vs.* 13.9, n.s.), and the remaining gap is in the answers (accuracy 21.5 *vs.* 26.3*). Even the oracle window leaks 45%: with every frame inside the evidence, the model still often starts near 0 s. More evidence alone does not remove the prior; the two levers are complementary.

6.9. Analysis of the optional retrieval

The SigLIP 2 index does find the evidence: over the seconds before each question, the AUROC of evidence seconds against the rest is 0.663, and the top-ranked second lies in the gold interval for 63% of questions (chance 36%) and 41% of location questions (chance 13.5%). Retrieval-guided frames put 40% of the memory inside the evidence, against 36% for uniform frames. This helps random tokens ($+3.8^*$ mIoU) but adds only $+0.8$ to $+0.9$ (n.s.) once tokens are chosen by the question, because both stages focus the memory on the question. Subtracting a question-free prior from SigLIP's score does not help (AUROC 0.635–0.666), as expected for a retriever trained with a contrastive loss, whose score already estimates pointwise mutual information [42]. Re-weighting the token budget of uniform frames by the same relevance adds $+1.3$ (n.s.): which frames are encoded matters more than how the budget is split among fixed frames.

6.10. What did not work

Table 24 collects the ideas that gave little or nothing. They share a lesson: at a fixed budget, redistributing evidence (zooming, re-weighting frames, unequal quotas) is close to its ceiling, and only removing the prior where it acts (PCD) or guaranteeing coverage (the equal quota) gives large gains.

6.11. Where answers still fail

Table 25 shows that PCD works where it was aimed: intervals entirely before the gold one, the signature of the prior, fall from the largest error type to less than a fifth of answers. The remaining failures point to four limits. (i) *Answers, not intervals*: twice as many answers are well grounded but wrong as grounded and right; counting answers do not use the events the model itself listed, and an IoU of 0.5 still allows a two-fold duration error. (ii) *Recent events*: 81% of the intervals that still end before the gold one have evidence in the second half of the window, where an earlier similar event is picked; GQ@0.5 is 4.4 when the evidence lies in the last third of the window. (iii) *Sparse evidence*: with fewer than 4 of the 64 frames inside the evidence, GQ@0.5 is 1–2% for any method, against 15% with 16 or more; no decoding rule recovers evidence that is not in the input. (iv) *Label ambiguity*: on the 98 questions whose annotators disagree (mean pairwise IoU < 0.6), every method scores near zero.

7. Limitations

Answer format. The main tables use a reasoning-style timeline prompt. On single-moment grounding it costs

Table 21. **Budget sweep at 64 frames** (mIoU). The second row gives the number of kept tokens per memory.

	S-EMBER			QVHighlights			ActivityNet		
Budget	5%	10%	20%	5%	10%	20%	10%	30%	50%
Kept tokens	∼1,070	∼2,140	∼4,280	∼260	∼520	∼1,040	120–300	400–900	600–1,500
Random	23.9	25.6	27.4		26.9		17.8	22.9	23.5
SemVID	20.9	23.3	26.5		24.5		21.2	22.8	22.3
R + C (no PCD)	34.6	32.5	31.0		34.5		11.3	25.4	25.0
SemVID + PCD	31.9	34.8	35.3		34.8		**26.8**	30.2	31.4
Random + PCD	34.5	35.0	36.5		37.8		21.5	31.4	32.2
CoCo	**39.3**	**39.3**	**36.8**	9.7	**48.8**	49.4	11.6	**34.7**	**34.4**

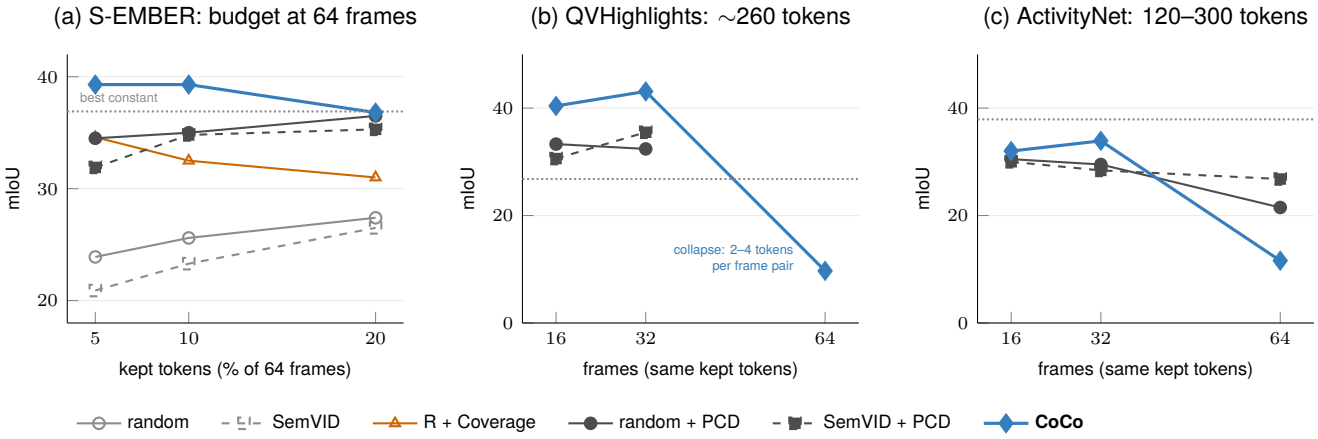


Figure 5. **Budget and how to spend it. (a)** At 64 frames, question-driven selection improves as the budget shrinks while random selection and SemVID degrade; CoCo leads at every budget that leaves at least about eight tokens per frame pair, and collapses below it (QVHighlights 5%, ActivityNet 10%). **(b, c)** At a fixed, very small number of kept tokens, 64 frames leave 2–4 tokens per frame pair and CoCo collapses; spending the same tokens on 32 frames recovers it (QVHighlights 9.7 → 43.1, ActivityNet 11.6 → 33.9). Dotted grey: best video-free constant.

the uncompressed model about 30 mIoU against the standard one-line prompt, and with it Qwen3-VL-8B stays below the best constant interval on ActivityNet Captions and Charades-STA. Our gains over the uncompressed model on QVHighlights, ActivityNet and Charades-STA (Tab. 4) are therefore relative to this format; under the one-line prompt the comparison is Tab. 9, and the full TimeLens-Bench sweep is in progress. On S-EMBER, CoCo beats the uncompressed model when both answer in the timeline format; the uncompressed model under the official direct-answer prompt is still to be reported. **[TODO: S-EMBER without pruning under the direct-answer and one-line prompts.]**

Where the prior is measured. Every measurement of the video-free prior (Sec. 3.3) uses the timeline format. PCD has no effect with a one-line answer, so the prior may decide few of the times written in that format, or none; we have not yet measured the no-video answers in it.

Minimum tokens per frame. Equal quotas fail when a frame pair keeps only 2–4 tokens. Spending the budget on fewer frames avoids this (Sec. 6.7), but the rule we use (at least about eight tokens per frame pair) was derived from the same results and still needs held-out validation.

Short, densely sampled videos. When uniform frames are less than a second apart (Charades-STA) or 64 frames already cover a 40 s video (NExT-GQA), question-driven selection keeps the answers but grounds no better than random selection.

Not a streaming memory. The memory is rebuilt per question from the frames before the question. It is causal, but the full memory is prefilled once per question, so peak memory and time-to-first-token are those of the uncompressed model: CoCo reduces the memory the decoder reads, not the cost of building it. PCD also decodes a second, prompt-only row at every step. **[TODO: cost table; amortisation over several questions per video; write-once memory re-ranked at question time.]**

Scope of the evaluation. Our two models come from one family: TimeLens-8B is fine-tuned from Qwen3-VL-

Table 22. **Spending a fixed token budget on fewer frames** (mIoU / R@0.5, all with PCD). Within each block every row keeps the same number of tokens: F frames at keep ratio $\rho \cdot 64/F$. Best and second best per row.

Dataset (kept tokens)	Frames × keep ratio	Random + PCD	SemVID + PCD	CoCo
QVHighlights (∼260)	64 × 5%	–	–	9.7 / 6.6
	32 × 10%	32.4 / 29.2	35.5 / 33.6	**43.1 / 46.6**
	16 × 20%	33.3 / 32.4	30.6 / 28.0	**40.4 / 41.0**
ActivityNet (120–300)	64 × 10%	21.5 / 18.2	**26.8 / 23.6**	11.6 / 7.6
	32 × 20%	29.5 / 26.8	28.4 / 25.8	**33.9 / 32.0**
	16 × 40%	30.5 / 26.8	30.0 / 27.6	**32.0 / 29.8**
Charades-STA (∼10% of 64)	64 × 10%	**26.2 / 24.0**	21.7 / 19.2	23.3 / 21.2
	32 × 20%	**24.6 / 22.0**	20.8 / 16.0	23.6 / 21.0[§]

[§] Coverage on uniform frames; retrieval is gated off on Charades-STA.

Table 23. **Upper bounds that use the gold interval** (S-EMBER, greedy unless noted; earlier runs: PCD without its answer reserve). Oracle selector: the gold frames' tokens are kept first. Oracle window: all 64 frames inside the gold interval.

Memory	Acc.	mIoU	R@0.5	GQ	Leak
No video	10.5	7.9	4.8	0.8	88%
Random 10%	15.2	25.8	24.2	7.2	30%
Random 10% + PCD	16.2	32.6	33.1	10.1	**11%**
Oracle selector 10%	17.5	27.9	25.7	8.4	33%
Gold-window zoom 10%	14.5	28.2	26.1	7.8	25%
All tokens	18.5	28.0	26.7	8.6	33%
All tokens + PCD	21.5	32.7	34.1	13.3	13%
Oracle window	**26.3**	**35.3**	**36.2**	**13.9**	45%

Table 24. **Negative results** (S-EMBER; ΔmIoU against the stated reference).

Idea	Reference	ΔmIoU
VCD-style noised-frame contrast	greedy	+0.5
Permuted-timestamp contrast	greedy	+0.9
Evidence-against decoding rule	PCD	−5.6*
Attention gain while decoding	greedy	+1.3
Bias stated in the instruction	greedy	−0.1
Relevance-guided frame zoom	greedy	−1.5
Zoom inside the gold window	greedy	+2.3
Unequal per-frame quotas (5 variants)	equal quota	−0.4 to +0.5
Contrastive token score	question top-k	−1.2 to +0.5
Prior-subtracted retrieval	SigLIP	AUROC ≤ +0.003
128 frames at the same budget	64 frames	+1.2 / −3.6*

8B, and both read time from text timestamps. Most diagnostic results use 475–500-question subsets, and answers are judged by a local model with the official prompt. Every hyperparameter was chosen on the S-EMBER questions; the other datasets were run once with the frozen configuration, but a held-out S-EMBER split is still to be evaluated. **[TODO: a second model family; held-out S-EMBER split.]**

Table 25. **Error types** on S-EMBER, pooled over eight memories, greedy *vs*. PCD (earlier runs: PCD without its answer reserve).

Outcome	Greedy	PCD	Δ
IoU ≥ 0.5, answer correct	6.1%	9.9%	+3.8*
IoU ≥ 0.5, answer wrong	16.3%	20.7%	+4.4*
Partial overlap (IoU < 0.5)	17.0%	15.7%	−1.3
Inside gold, too short	22.2%	18.5%	−3.7*
Entirely *before* gold	33.1%	18.9%	−14.2*
Entirely after gold	2.7%	5.7%	+3.0*
Covers gold, too long	2.2%	2.5%	+0.3
No answer / unparsed	0.5%	8.1%	+7.6*

8. Conclusion

Pruning for QA keeps what is salient; grounding needs what is everywhere. A video LLM reads time from anchors that survive only with their frames, so pruners built for QA, which empty unremarkable frames, fall below random selection on grounding. Keeping one anchor in every frame, filled with what the question attends to, is enough to match the uncompressed model with a quarter of the visual tokens under the standard grounding prompt, on a general and a grounding-tuned model. When the answer lists what was seen before it states the interval, as grounded QA does, a second failure appears: where the memory is silent, a video-free prior writes the times. Contrasting each time value with the model's own no-video prediction removes it, and with a tenth of the tokens CoCo then grounds better than the uncompressed model answering in the same format. The answer format itself turns out to be a first-order variable, worth about 30 mIoU on single-moment grounding, and with video-free floors it belongs in every evaluation of compressed video memories.

References

[1] [authors to verify]. AdaTP: Attention-debiased token prun-

ing for video large language models. *arXiv preprint arXiv:2505.20100*, 2025. 2, 3

[2] [authors to verify]. FOCUS: Efficient keyframe selection for long video understanding. *arXiv preprint arXiv:2510.27280*, 2025. 3

[3] [authors to verify]. InfiniPot-V: Memory-constrained KV cache compression for streaming video understanding. *arXiv preprint arXiv:2506.15745*, 2025. 3

[4] [authors to verify]. Language-guided temporal token pruning for efficient VideoLLM processing. In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2025. [verify]. 3

[5] [authors to verify]. MDP3: A training-free approach for list-wise frame selection in video-LLMs. *arXiv preprint arXiv:2501.02885*, 2025. 3

[6] [authors to verify]. Q-Frame: Query-aware frame selection and multi-resolution adaptation for video-LLMs. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025. [verify]. 3

[7] [authors to verify]. SEASON. *arXiv preprint arXiv:2512.04643*, 2025. [verify title]. 3

[8] [authors to verify]. StreamMem: Query-agnostic KV cache memory for streaming video understanding. *arXiv preprint arXiv:2508.15717*, 2025. 3

[9] [authors to verify]. Your VLM already knows when. *arXiv preprint arXiv:2608.08315*, 2026. [verify title]. 3

[10] [authors to verify]. HERMES. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2026. [verify title]. 3, 6

[11] [Authors to verify]. MLLMs know when before speaking. *arXiv preprint arXiv:2605.21954*, 2026. [verify title]. 3

[12] [authors to verify]. Keyframes and key tokens selection for efficient training-free video LLMs. *arXiv preprint arXiv:2602.03615*, 2026. 2, 3

[13] [authors to verify]. REZE. *arXiv preprint arXiv:2608.04480*, 2026. [verify title]. 3

[14] [authors to verify]. SemVID. In *European Conference on Computer Vision (ECCV)*, 2026. arXiv:2603.05663; [verify title]. 2, 3, 7, 9

[15] Zefan Cai et al. PyramidKV: Dynamic KV cache compression based on pyramidal information funneling. *arXiv preprint arXiv:2406.02069*, 2024. 3

[16] Shangzhe Di et al. Streaming video question-answering with in-context video KV-cache retrieval. In *International Conference on Learning Representations (ICLR)*, 2025. 3

[17] Ziyang Fan et al. FlashVID: Training-free tree-based spatiotemporal token merging for video large language models. In *International Conference on Learning Representations (ICLR)*, 2026. arXiv:2602.08024; [verify]. 2, 3, 7, 9

[18] Alessandro Favero, Luca Zancato, Matthew Trager, Siddharth Choudhary, Pramuditha Perera, Alessandro Achille, Ashwin Swaminathan, and Stefano Soatto. Multi-modal hallucination control by visual information grounding. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 2, 3

[19] Yuan Feng et al. Ada-KV: Optimizing KV cache eviction by adaptive budget allocation for efficient LLM inference. *arXiv preprint arXiv:2407.11550*, 2024. 3

[20] Tianyu Fu et al. FrameFusion: Combining similarity and importance for video token reduction on large visual language models. *arXiv preprint arXiv:2501.01986*, 2025. [verify]. 3

[21] Jiyang Gao, Chen Sun, Zhenheng Yang, and Ram Nevatia. TALL: Temporal activity localization via language query. In *IEEE International Conference on Computer Vision (ICCV)*, 2017. 7

[22] Ari Holtzman, Peter West, Vered Shwartz, Yejin Choi, and Luke Zettlemoyer. Surface form competition: Why the highest probability answer isn't always right. *arXiv preprint arXiv:2104.08315*, 2021. 3

[23] Bin Huang, Xin Wang, Hong Chen, Zihan Song, and Wenwu Zhu. VTimeLLM: Empower LLM to grasp video moments. *arXiv preprint arXiv:2311.18445*, 2023. 3

[24] De-An Huang, Shijia Liao, Subhashree Radhakrishnan, Hongxu Yin, Pavlo Molchanov, Zhiding Yu, and Jan Kautz. LITA: Language instructed temporal-localization assistant. *arXiv preprint arXiv:2403.19046*, 2024. 3

[25] Xiaohu Huang, Hao Zhou, and Kai Han. PruneVid: Visual token pruning for efficient video large language models. In *Findings of the Association for Computational Linguistics (ACL)*, 2025. 2, 3

[26] Ranjay Krishna, Kenji Hata, Frederic Ren, Li Fei-Fei, and Juan Carlos Niebles. Dense-captioning events in videos. In *IEEE International Conference on Computer Vision (ICCV)*, 2017. 7

[27] Jie Lei, Tamara L. Berg, and Mohit Bansal. QVHighlights: Detecting moments and highlights in videos via natural language queries. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021. 7

[28] Sicong Leng, Hang Zhang, Guanzheng Chen, Xin Li, Shijian Lu, Chunyan Miao, and Lidong Bing. Mitigating object hallucinations in large vision-language models through visual contrastive decoding. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 2, 3, 7, 13

[29] Xiang Lisa Li, Ari Holtzman, Daniel Fried, Percy Liang, Jason Eisner, Tatsunori Hashimoto, Luke Zettlemoyer, and Mike Lewis. Contrastive decoding: Open-ended text generation as optimization. In *ACL*, 2023. 2, 3

[30] Yuhong Li, Yingbing Huang, Bowen Yang, Bharat Venkitesh, Acyr Locatelli, Hanchen Ye, Tianle Cai, Patrick Lewis, and Deming Chen. SnapKV: LLM knows what you are looking for before generation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. 2, 3

[31] Shi Liu, Kecheng Zheng, and Wei Chen. Paying more attention to image: A training-free method for alleviating hallucination in LVLMs. In *European Conference on Computer Vision (ECCV)*, 2024. 3

[32] Shuming Liu et al. BOLT: Boost large vision-language model without training for long-form video understanding. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. arXiv:2503.21483. 3, 7

[33] Xuyang Liu et al. Video compression commander: Plug-and-play inference acceleration for video large language models. In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2025. arXiv:2505.14454. 2, 3, 7, 9

[34] Mayu Otani, Yuta Nakashima, Esa Rahtu, and Janne Heikkilä. Uncovering hidden challenges in query-based video moment retrieval. In *British Machine Vision Conference (BMVC)*, 2020. 2, 4

[35] Qwen Team. Qwen3-VL technical report, 2025. [verify]. 2, 7

[36] Shuhuai Ren, Linli Yao, Shicheng Li, Xu Sun, and Lu Hou. TimeChat: A time-sensitive multimodal large language model for long video understanding. In *CVPR*, 2024. 3

[37] Guillaume Sanchez, Honglu Fan, Alexander Spangher, Elad Levi, Pawan Sasanka Ammanamanchi, and Stella Biderman. Stay on topic with classifier-free guidance. *arXiv preprint arXiv:2306.17806*, 2023. 2, 3

[38] Leqi Shen et al. FastVID: Dynamic density pruning for fast video large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. arXiv:2503.11187. 2, 3, 7, 9

[39] Weijia Shi, Xiaochuang Han, Mike Lewis, Yulia Tsvetkov, Luke Zettlemoyer, and Wen-tau Yih. Trusting your evidence: Hallucinate less with context-aware decoding. *arXiv preprint arXiv:2305.14739*, 2023. 2, 3

[40] Xi Tang et al. Adaptive keyframe sampling for long video understanding. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. arXiv:2502.21271. 3

[41] Keda Tao et al. DyCoke: Dynamic compression of tokens for fast video large language models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. [verify]. 2, 3

[42] Michael Tschannen et al. SigLIP 2: Multilingual vision-language encoders with improved semantic understanding, localization, and dense features. *arXiv preprint arXiv:2502.14786*, 2025. 7, 14

[43] Xintong Wang, Jingheng Pan, Liang Ding, and Chris Biemann. Mitigating hallucinations in large vision-language models with instruction contrastive decoding. In *Findings of the Association for Computational Linguistics (ACL)*, 2024. 3

[44] Xiaodong Wang, Xuanyi Zhao, Pedro Rodriguez, Devendra Singh Sachan, Barlas Oguz, Seungwhan Moon, Shang-Wen Li, Gargi Ghosh, Xin Dong, and Wen-tau Yih. S-EMBER: A large-scale benchmark for streaming egocentric memory retrieval. *arXiv preprint arXiv:2607.02689*, 2026. 7

[45] Yongliang Wu et al. Number it: Temporal grounding videos like flipping manga. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. [verify]. 3

[46] Junbin Xiao, Angela Yao, Yicong Li, and Tat-Seng Chua. Can i trust your answer? visually grounded video question answering. In *CVPR*, 2024. 7, 8

[47] Jun Zhang, Teng Wang, Yuying Ge, Yixiao Ge, Xinhao Li, Ying Shan, and Limin Wang. Timelens: Rethinking video temporal grounding with multimodal llms. *arXiv preprint arXiv:2512.14698*, 2025. 3, 7, 8

[48] Zhenyu Zhang, Ying Sheng, Tianyi Zhou, Tianlong Chen, Lianmin Zheng, Ruisi Cai, Zhao Song, Yuandong Tian, Christopher Ré, Clark Barrett, Zhangyang Wang, and Beidi Chen. H2O: Heavy-hitter oracle for efficient generative inference of large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. 3

[49] Tony Z. Zhao, Eric Wallace, Shi Feng, Dan Klein, and Sameer Singh. Calibrate before use: Improving few-shot performance of language models. In *ICML*, 2021. 2, 3

A. Parsing times written as minutes:seconds

On videos longer than a minute, the model sometimes writes times as minutes:seconds, *e.g.* `Time: [1:45, 2:07]`, which the official S-EMBER parser scores as missing. We convert `h:mm:ss` and `m:ss` to seconds for every method alike. Method rankings are unchanged with the official parser. The main numbers with the official parser are: QVHighlights, CoCo 44.6 / 50.2 (*vs.* 48.8 / 54.8); ActivityNet at 30%, 33.3 / 31.0 (*vs.* 34.7 / 32.4). **[TODO: full table with the official parser.]**

B. S-EMBER closed intervals

Two thirds of S-EMBER gold intervals end at, or close to, the question time, so a constant interval ending at q is a strong floor on the pooled set. Table 26 restricts the main comparison to the *closed* questions (gold end $< 0.8q$, 33% of the questions), where that shortcut does not exist; the floors are re-tuned on this subset.

Table 26. **S-EMBER closed intervals** (10%, mIoU / R@0.5; settings as in Tab. 4). Right: paired ΔmIoU of CoCo (*: 95% paired bootstrap interval excludes 0).

Method	mIoU / R@0.5	CoCo *vs.*
Best constant interval	*32.1 / 29.9*	+7.6*
Length rescaling of R + C	*32.3 / 33.1*	+7.5*
All tokens (no pruning)	30.0 / 27.4	
Random tokens	28.3 / 27.4	
Random + PCD	36.8 / 38.2	
SemVID + PCD	34.3 / 34.4	
Coverage only	33.0 / 35.7	
CoCo (Coverage + PCD)	**39.7 / 43.3**	–

C. Checks against published numbers

Before comparing pruning methods under the TimeLens-Bench protocol, we reproduce the published unpruned scores with the TimeLens evaluation script, and check that our own inference pipeline gives the same scores on the same queries (Tab. 27).

D. End-snapping on S-EMBER

Setting every predicted end to the question time q raises every S-EMBER method to 42–46 mIoU, more than the 39.3 of CoCo, because 67% of the gold intervals end within the last fifth of the window. On closed intervals, end-snapping *hurts* every method (to 22–24 mIoU) while CoCo keeps 39.7; on QVHighlights it lowers CoCo from 48.8 to 40.5. A decoder that decides whether an interval is still open at question time is a natural extension. **[TODO: open-interval decoding.]**

Table 27. **Reproduction of TimeLens-Bench** (mIoU, no pruning). Top: TimeLens script on the full sets *vs.* published. Bottom: our pipeline *vs.* the TimeLens script on the same queries (paired Δ).

Model	Dataset (queries)	Ours	Reference	Δ
Qwen3-VL-8B	Charades-STA (3,363)	48.5	48.3 (published)	
Qwen3-VL-8B	QVHighlights (1,541)	59.6	59.4 (published)	
Qwen3-VL-8B	ActivityNet (4,500)	47.1	46.8 (published)	
TimeLens-8B	Charades-STA (3,363)	55.4	55.2 (published)	
TimeLens-8B	QVHighlights (1,541)	65.7	65.5 (published)	
TimeLens-8B	ActivityNet (4,500)	53.2	53.2 (published)	
Qwen3-VL-8B	Charades-STA (400)	48.0	47.5 (script)	+0.5
Qwen3-VL-8B	ActivityNet (400)	47.1	45.7 (script)	+1.4
Qwen3-VL-8B	QVHighlights (513)	60.5	60.5 (script)	0.0
TimeLens-8B	QVHighlights (513)	67.1	66.1 (script)	+1.0

E. Reproduction

CoCo corresponds to the following overrides of our code (released with the paper):

```
model=qwen3_vl_8b experiment=sember_grounding_uniform run.uniform_num_frames=64
dataset.grounding_prompt=timeline run.keep_time_tokens=surviving run.max_new_tokens=384
run.batch_size=8 run.prune_score=core_topk run.offline_keep_ratio=0.1
run.contrastive_mode=blind run.contrastive_alpha=1.0 run.contrastive_adaptive=true
run.contrastive_scope=time run.answer_reserve=64
(optional retrieval) run.retrieval=frames_cont run.retrieval_min_gap=1.0
```

Floors and paired intervals are computed by `docs/paper/analysis/floors.py`.